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Frequency-dependent dielectric and magnetic properties of epoxy-based carbon nanotube composites containing inorganic fillers

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ABSTRACT

The frequency-dependent dielectric and magnetic properties of two- and three-phase epoxy composites with multi-walled carbon nanotubes (MWCNTs) and inorganic fillers of different morphologies were studied. Spherical Co_3O_4 nanoparticles, spherical Co microparticles, and disc-shaped Fe particles were used. The filler morphology strongly affects interfacial polarization and percolation threshold. Composites with Co_3O_4 show the highest dielectric permittivity ($\epsilon' \approx 900$, $\epsilon'' \approx 1200$) at 5–7 wt% MWCNT. AC conductivity was analyzed using Jonscher's power law and Dyre's model. The magnetic properties of the composite depend on the type of filler: the addition of Co and Fe particles increases μ' and μ'' in the composite.

KEYWORDS

inorganic fillers; multiwall carbon nanotubes; percolative composite; permeability; permittivity; polymers

The manuscript has not been published elsewhere, and it has not been submitted simultaneously for publication elsewhere.

1. Introduction

Polymer–matrix composites containing various carbon nanofillers have been extensively studied for applications in electromagnetic shielding, mechanical strain sensing (piezoresistive elements), electromagnetic interference (EMI) protection, and antistatic coatings [1–9]. Although the dielectric permittivity of such composites generally decreases in the microwave range, its value remains sufficiently high to render these materials attractive both for shielding and for efficient absorption of microwave radiation. Carbon nanostructures with a high aspect ratio (MWCNTs, graphite nanoplatelets (GNPs), carbon nanofibers, graphene, etc.) provide low percolation thresholds and a pronounced increase in the effective dielectric permittivity at filler contents above the percolation threshold [10–18].

A well-known drawback of composites with carbon nano-inclusions is the tendency of nanoparticles to form clusters of different sizes within the polymer matrix, so that

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the electromagnetic response is largely governed by cluster formation rather than by an ideal homogeneous dispersion. To control the microstructure and the electrodynamic response, additional fillers – metallic, inorganic, or magnetic particles – are introduced into the composite matrix [13–16]. In multiphase systems, the emergence of a synergy effect is of particular importance: the interaction between different fillers can reduce the effective percolation threshold and significantly improve the electromagnetic performance compared with single-filler systems [19]. Frequently, the second filler is chosen among inorganic particles with a dielectric permittivity much higher than that of the matrix (e.g. ferroelectric ceramics), which makes it possible to increase ϵ' while maintaining moderate dielectric losses [15,16].

For binary epoxy composites with MWCNTs and graphene/GNPs, dielectric and electrical characteristics have been systematically investigated over a broad frequency range. In epoxy/MWCNT systems, the real part of the dielectric permittivity ϵ' in the MHz–GHz range increases from a few units to values on the order of several tens as the concentration approaches the percolation threshold, accompanied by a sharp growth of dielectric losses and AC conductivity [16,17]. Epoxy/GNPs composites exhibit a similar tendency: ϵ' increases with increasing nanoplatelet content and reaches values of several tens near percolation limit, while both the conductivity and the microwave absorption are enhanced [6–8]. These results indicate that the dominant contribution to the increase in ϵ' in nanocarbon/epoxy systems is associated with the formation of conductive nanocarbon clusters, whereas inorganic and/or magnetic additives act primarily as modifiers of the percolation threshold, loss level, and frequency dispersion of the dielectric response [20].

Dielectric permittivity is one of the key physical parameters governing the response of a material to an applied electric field [21]. The most pronounced changes in ϵ' as well as ϵ'' upon the addition of fillers to a polymer matrix are observed for nanocarbon particles with a high aspect ratio [22]. In such systems, ϵ' increases with increasing content of the conductive filler up to concentrations close to the percolation threshold; beyond this point the composite becomes electrically conductive and the electrical losses (ϵ'' and $\tan \delta$) rise sharply [16,17,23]. For CNT–PVDF composites with a CNT content of ~ 0.12 vol.%, ϵ' can reach values on the order of 10^3 , while $\tan \delta$ may attain values of $\sim 10^2$ [24,25]. A number of studies have demonstrated that, under specific conditions, negative values of the effective dielectric permittivity ($\epsilon' < 0$) can appear in such percolative systems, which is interpreted as a manifestation of metamaterial-like behavior in percolative metacomposites [26–32]. In the present work, these effects are mentioned only as an important reference; the main emphasis is placed on the conditions required to obtain high ϵ' and on the characteristic dispersion of $\epsilon'(f)$, $\epsilon''(f)$ and $\sigma(f)$ in the MHz range.

Epoxy composites with MWCNTs and magnesium oxide (MgO), studied in Bertasius *et al.* [16], provide a representative example of multicomponent polymer systems exhibiting interfacial polarization. All samples show a typical decrease in ϵ' with increasing frequency due to the gradual loss of the ability of dipoles to follow a rapidly alternating electric field. The highest ϵ' values in the low-frequency region are observed for samples containing 1 vol% MgO, which indicates efficient formation of polarized interfacial regions. At 3 vol% MgO, ϵ' is markedly reduced over the entire frequency range, which

was attributed to MgO nanoparticle agglomeration that suppresses a uniform interfacial interaction with MWCNTs. The electrical conductivity $\sigma(f)$ also exhibits characteristic percolative behavior: a low-frequency DC plateau is followed by a power-law increase with frequency; optimal MgO concentrations yield a synergistic improvement of conductivity, whereas an excess of dielectric filler disrupts the conductive network [16].

Against this background, multiphase epoxy composites with nanocarbon fillers are of particular interest, since the introduction of additional inorganic (in particular, magnetic) particles offers a route to control the formation of conductive clusters, the strength of interfacial polarization, and the level of dielectric losses in the radio-frequency range.

Recent studies have also demonstrated the important role of filler concentration, morphology, and interfacial interactions in determining the dielectric behavior and structure–property relationships of polymer-based composite systems over different frequency ranges [33–35].

Despite numerous studies devoted to polymer composites containing carbon nanotubes and inorganic fillers, the combined influence of filler morphology, characteristic size, and spatial distribution on the dielectric and conductive response of MWCNT/epoxy composites remains insufficiently systematized, especially in the radio-frequency range. In particular, a direct comparative analysis of composites containing nanosized spherical Co_3O_4 particles, spherical Co microparticles, and disk-like Fe microparticles at the same inorganic-filler loading has not been sufficiently addressed.

The novelty of this work lies in the systematic comparative investigation of epoxy-based MWCNT composites containing inorganic fillers with substantially different morphologies and characteristic sizes under identical experimental conditions. Such an approach makes it possible to establish structure–property relationships between the morphology of the secondary filler and the processes of conductive network formation, interfacial polarization, dielectric response, AC/DC conductivity, and magnetic permeability of the composites.

The present work aims to investigate how the morphology of secondary inorganic fillers influences the electrodynamic response of MWCNT/epoxy composites in the frequency range 1–500 MHz. Particular attention is paid to the influence of filler morphology on conductive pathway formation, dielectric losses, and magnetic response in composites with a high density of interfacial boundaries.

2. Experimental section

2.1. Materials and fabrication

The polymer composite materials under study were prepared using an epoxy matrix Larit285 (L285) (Lange & Ritter GmbH, Gerlingen, Germany), whose viscosity is 600–1000 mPa·s at 25 °C. Curing was carried out using hardener H285 at a mass ratio of 100:40 (epoxy resin: hardener) at room temperature.

Two main types of composites were fabricated. The first type consisted of two-phase systems containing 1–10 wt% multi-walled carbon nanotubes (MWCNTs, Nanocyl NC7000, purity 95%) dispersed in the L285 epoxy matrix (x%MWCNT-L285). The second type comprised three-phase systems containing 2–10 wt%MWCNTs, epoxy resin

L285, and an additional inorganic filler at a fixed content of 30 wt% (and 60 wt% only with L285) (cobalt [Co], carbonyl iron [Fe], or cobalt oxide [Co_3O_4]) (x%MWCNT/Co(Fe, Co_3O_4)-L285). Neat epoxy resin L285 without fillers was used as a reference sample.

Several types of fillers with markedly different morphologies and characteristic dimensions were employed (Fig. 1). The primary nanocarbon component consisted of MWCNTs with a cylindrical morphology, an average length of approximately $10\text{ }\mu\text{m}$, and diameters in the range of $10\text{--}30\text{ nm}$. This results in a high aspect ratio on the order of $330\text{--}1000$, characteristic of strongly elongated particles.

Cobalt (Co) particles are approximately spherical, with diameters below $2\text{ }\mu\text{m}$ and an aspect ratio of ~ 1.2 , typical of nearly isotropic inclusions (Fig. 1b). Carbonyl iron (Fe) particles exhibit a disc-like morphology with in-plane dimensions of $3\text{--}7\text{ }\mu\text{m}$ and a thickness of $0.6\text{--}2\text{ }\mu\text{m}$, corresponding to an aspect ratio of about three and a moderately flattened geometry. Carbonyl Fe powder was supplied by Ukrspav Inc (Fig. 1c). Cobalt oxide (Co_3O_4) forms spherical nanoparticles with diameters up to 50 nm and a similar

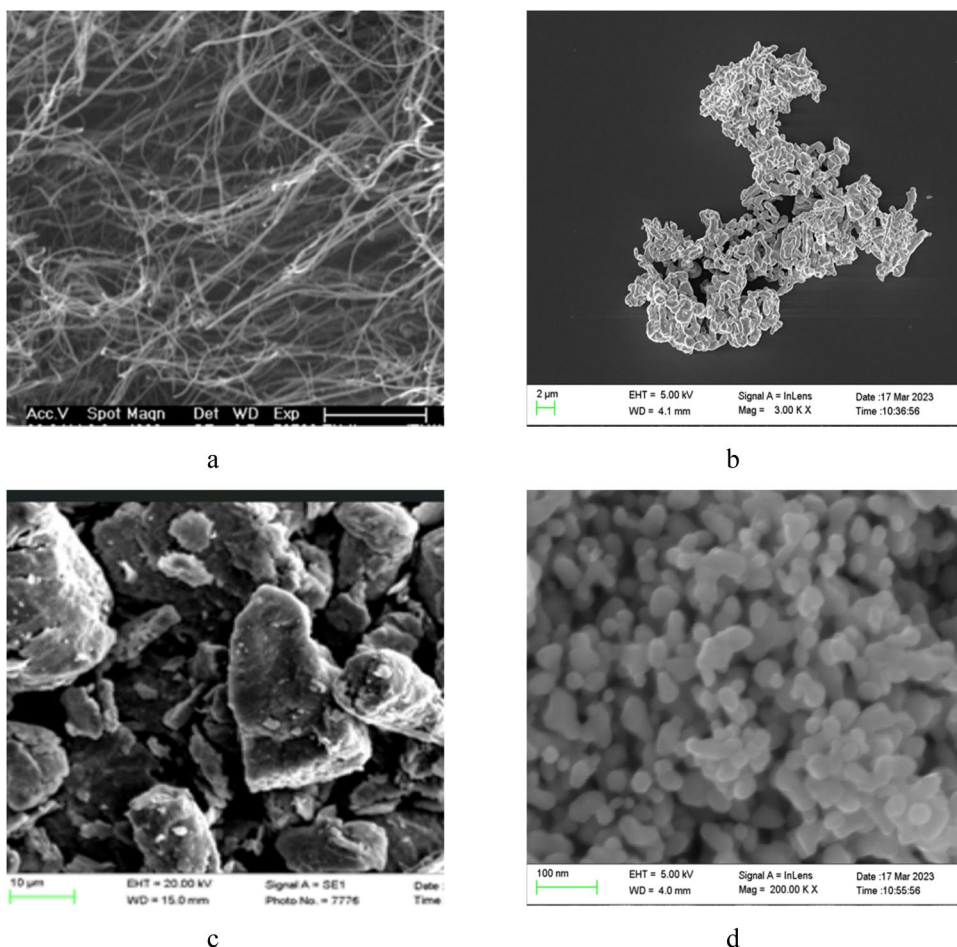
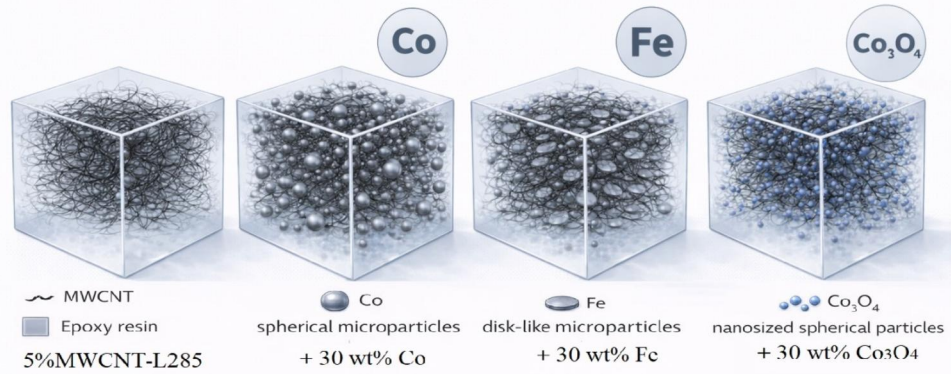


Figure 1. SEM images of filler particles: (a) MWCNTs, (b) cobalt (Co), (c) carbonyl iron (Fe), and (d) cobalt oxide (Co_3O_4).

Table 1. Particle size, specific surface area S_m and intrinsic conductivity σ_f of fillers

Filler type	MWCNT	Co	Fe	Co ₃ O ₄
Particles size	Cylindrical, length – 10 μm , diameter – 10–30 nm	Spherical, diameter – – 2 μm	Disk-like, Lateral size – 3–7 μm , thickness – 0.6–2 μm	Spherical, diameter – 50 nm
S_m , m ² /g	~ 100	~ 0.34	~0.20	~ 20
σ_f S/m	6×10^4 [36]	$\sim 1.6 \times 10^7$ [37]	$\sim 1 \times 10^7$ [37]	0.1 [38]

**Figure 2.** Schematic illustration of the distribution of carbon nanotubes within the polymer matrix containing inorganic fillers with different shapes and sizes.

aspect ratio (~ 1.2). Table 1 presents the characteristic particle size, intrinsic electrical conductivity reported in the literature [36–38], and the specific surface area of the fillers. The specific surface area values were estimated from the particle dimensions and morphology.

The combination of highly anisotropic MWCNTs with quasi-spherical Co and Co₃O₄ particles and disc-like Fe inclusions provides a broad range of morphological characteristics and interfacial geometries, which is particularly suitable for analyzing the influence of filler shape and size on the dielectric and electrodynamic properties of epoxy composites.

Series of three-phase epoxy composites x%MWCNT/Co-L285, x%MWCNT/Fe-L285, and x%MWCNT/Co₃O₄-L285 were prepared following the same technological protocol, varying only the type of inorganic filler at a fixed content of 30 wt% while adjusting the concentration of carbon nanotubes (x) in the range of 2–10 wt% (Fig. 2).

The polymer composite materials were fabricated by mechanical mixing of the components followed by ultrasonic treatment. The required amount of epoxy resin L285 was weighed into a test tube, after which the target mass fractions of MWCNTs (2–10 wt%), inorganic filler (30 wt%), and hardener H285 (40 wt% relative to the mass of epoxy resin L285) were calculated [39]. The calculated amounts of MWCNTs and the corresponding inorganic filler were added to the L285 resin and thoroughly premixed.

Subsequently, ultrasonic dispersion was performed for 2 h in a BAKU ultrasonic bath operating at a frequency of 40 kHz and a power of 50 W, which helped to minimize

filler aggregation and improve dispersion within the epoxy matrix. At the final stage, hardener H285 was added, the mixture was thoroughly mixed, and the composite was poured into molds. Curing was carried out at room temperature for 24 h.

2.2. Characterizations and measurements

To obtain information on the phase composition and morphology of Co, Fe, and Co_3O_4 powders, scanning electron microscopy (SEM) studies were performed using a LEO 1550 Gemini field-emission scanning electron microscope (Zeiss, Jena, Germany).

The complex dielectric and magnetic properties of the samples in the frequency range of 1–500 MHz were measured using a Keysight E4991B impedance analyzer equipped with two different measurement fixtures, specifically designed for dielectric and magnetic characterization, respectively. According to the manufacturer specifications, the instrument provides a basic measurement accuracy of up to $\pm 0.65\%$ over a wide impedance range and supports open/short/load calibration procedures.

The complex dielectric permittivity of the samples was determined using a Keysight 16453 A dielectric test fixture, which enables direct determination of the complex dielectric permittivity $\epsilon^* = \epsilon' - i\epsilon''$ from the measured complex impedance. Disc-shaped samples compatible with the geometry of the 16453 A fixture were used; the sample diameter did not exceed 20 mm and the thickness was below 8.5 mm, ensuring complete coverage of the electrode area and proper positioning inside the fixture.

The complex magnetic permeability of the samples was measured on the same impedance analyzer using a Keysight 16454 A magnetic material test fixture, which allows determination of the real and imaginary parts of the magnetic permeability $\mu^* = \mu' - i\mu''$ based on inductive impedance measurements in the radio-frequency range.

Ring-shaped (toroidal) samples compatible with the geometry of the 16454 A fixture were used for magnetic measurements. The outer diameter of the samples did not exceed 20 mm, the inner diameter was at least 5 mm, and the thickness was below 8.5 mm, in accordance with the fixture specifications.

The measurement principle is based on inductive impedance measurements of a toroidal sample inserted into the fixture, which forms an ideal single-turn inductor with negligible magnetic flux leakage. The complex magnetic permeability $\mu^* = \mu' - i\mu''$ was calculated from the frequency-dependent inductive impedance response of the sample. Here, the real part μ' characterizes the energy stored in the magnetic material, while the imaginary part μ'' corresponds to magnetic losses in the alternating electromagnetic field.

Prior to each series of measurements, the impedance analyzer was calibrated using the built-in software calibration procedure and the reference standards supplied with the corresponding fixtures. Before measurements, the electrode and contact surfaces were carefully cleaned, and the samples were accurately positioned within the fixtures to ensure reproducible electrical contact and to minimize edge and positioning effects.

All measurements were performed at room temperature. As a result, the frequency dependences of the real and imaginary parts of the dielectric permittivity, $\epsilon'(f)$ and $\epsilon''(f)$, as well as the real and imaginary parts of the magnetic permeability, $\mu'(f)$ and $\mu''(f)$, were obtained in the frequency range of 1–500 MHz.

The selected frequency range of 1–500 MHz is particularly suitable for analyzing interfacial polarization processes, conductive losses, and percolation-related dielectric behavior in polymer composites. These mechanisms play a critical role in the electrodynamic response of nanocarbon-filled systems and are important for further development of electromagnetic shielding materials over broader frequency ranges, including the GHz region.

The electrical resistance of the composite samples was measured at room temperature under direct current (DC) conditions using a standard two-probe technique. Measurements were performed on rectangular prism-shaped specimens with dimensions of $3 \times 3 \times 7 \text{ mm}^3$.

3. Results and discussion

3.1. Frequency dependence of the real (ϵ') and imaginary (ϵ'')

Parts of Complex Permittivity in Three-Component MWCNT/L285 Epoxy Composites Containing Co, Fe, and Co_3O_4 (1–500 MHz). For polymer composites filled with conductive particles, the relative complex permittivity is conveniently expressed as follows [40,41]:

$$\epsilon_r^* = \epsilon_r' - i\epsilon_r'', \quad (1)$$

$$\epsilon_r'' = \epsilon_{rD}'' + \epsilon_{rP}'' + \epsilon_{rC}'', \quad (2)$$

where ϵ_r' is the real part of the permittivity, characterizing the energy stored in the material, and ϵ_r'' is the imaginary part, associated with dielectric losses, ϵ_{rD}'' corresponds to dipolar losses, ϵ_{rP}'' represents interfacial (Maxwell – Wagner) polarization losses, and ϵ_{rC}'' accounts for conductive losses.

Figure 3 illustrates the frequency dependence of the real (ϵ' , a) and imaginary (ϵ'' , b) parts of the complex dielectric permittivity for two-phase x%MWCNT–L285 composites with nanotube content ranging from 1 to 10 wt%. As shown in Fig. 3a, all samples exhibit a typical decrease in ϵ' with increasing frequency, which is associated with the reduced efficiency of polarization processes in the polymer matrix [42]. For samples containing 1–3 wt%MWCNTs, the real permittivity ϵ' increases from ~ 15 (1 wt%) to ~ 22 (2 wt%) and up to ~ 45 (3 wt%) at low frequencies, indicating the gradual formation of polarized MWCNTs clusters without a continuous conductive network. These values are characteristic of dielectric systems with localized conductive regions, where interfacial polarization is still not fully developed.

For comparison, the neat epoxy matrix L285 exhibits a constant ϵ' of ~ 3.8 (at 10^6 Hz) over the entire frequency range, without pronounced dispersion, indicating the absence of conductive or polarizable inclusions. At 5 wt%MWCNTs, the real permittivity of the composite at low frequencies ($f \sim 1 \text{ MHz}$) reaches ~ 130 , suggesting proximity to the percolation threshold [42]. At this stage, a dense but not yet fully conductive nanotube network is formed, enabling strong interfacial polarization at phase boundaries. The system stores maximal polarization charge without significant screening due to conduction.

In contrast, at 10 wt%MWCNTs, ϵ' decreases to ~ 80 , which reflects a transition to a semiconducting regime: the system loses its ability to store electric energy and instead

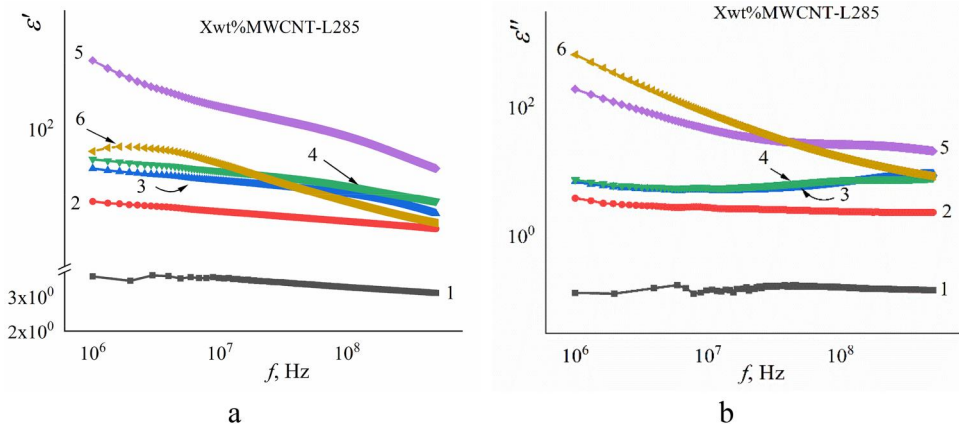


Figure 3. Frequency dependences of the real (ϵ') (a) and imaginary (ϵ'') (b) parts of the complex permittivity of MWCNT-L285 epoxy composites with MWCNT contents of 1–10 wt%. 1–100 wt% L285, 2–1 wt% MWCNT-L285, 3–2 wt% MWCNT-L285, 4–3 wt% MWCNT-L285, 5–5 wt% MWCNT-L285, 6–10 wt% MWCNT-L285.

generates losses. This behavior is typical for conductive composites above the percolation threshold [41].

The imaginary part of the permittivity, $\epsilon''(f)$, shown in Fig. 3b, increases gradually with rising MWCNT concentration. For the sample containing 1 wt% MWCNTs, ϵ'' values remain low and exhibit virtually no frequency dispersion. In samples with 2 and 3 wt% MWCNTs, a moderate increase in ϵ'' is observed at frequencies below 10^7 Hz. The 5 wt% MWCNTs sample shows a more pronounced low-frequency increase in ϵ'' , although no distinct peak is registered within the measured range. This behavior suggests a transition between relaxation losses and the onset of conduction-dominated contributions.

Specifically, for samples with 1–5 wt% MWCNTs, broad ϵ'' maxima appear in the 10^6 – 10^7 Hz region, which may be attributed to Maxwell–Wagner interfacial polarization mechanisms [43]. At 10 wt% MWCNTs, ϵ'' increases steeply up to ~ 600 at $f \approx 10^6$ Hz and lower, without a discernible peak. This response is typical of systems where dielectric losses are governed by conductivity (ϵ''_{rC}) in accordance with Equation (2). The overall behavior is consistent with a dielectric-to-conductive transition, as also reported in the literature for CNT/epoxy composites [44,45].

Thus, the comprehensive analysis of the frequency dependence of ϵ' and ϵ'' confirms that the dominant mechanisms of the dielectric response in MWCNT–L285 composites evolve with increasing nanotube content: from isolated interfacial polarization at low concentrations to a conduction-driven regime at 10 wt% MWCNTs. Although no distinct ϵ'' maxima were observed within the investigated frequency range, the increase in ϵ'' at lower frequencies suggests that such peaks may occur below 10^6 Hz. These findings are in good agreement with literature data for CNT-filled polymers, where the dielectric behavior (ϵ' and ϵ'') is known to depend strongly on the degree of aggregation, spatial distribution of the filler, and proximity to the percolation threshold [44,45].

Figure 4 presents the frequency dependence of the real (ϵ' , a) and imaginary (ϵ'' , b) parts of the complex dielectric permittivity for three-phase epoxy composites of the type

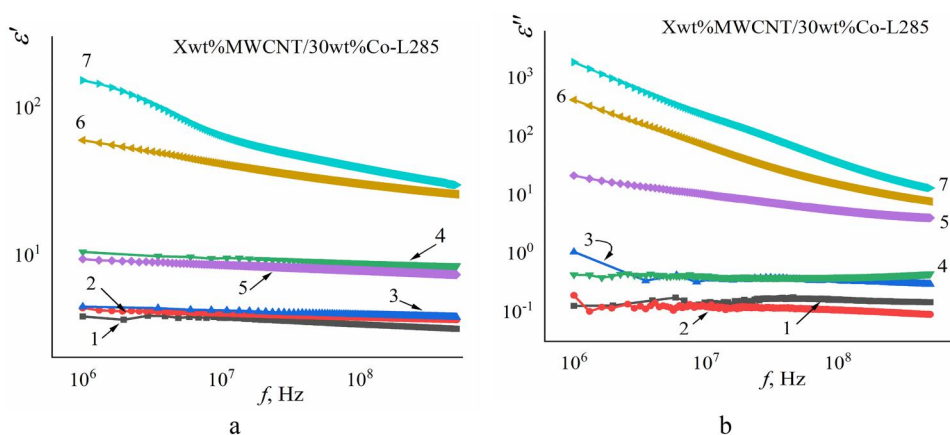


Figure 4. Frequency dependences of ε' (a) and ε'' (b) for MWCNT/30 wt%Co-L285 composites at MWCNT loadings of 2–10 wt%, as well as for the 60 wt% Co-L285 composite without MWCNTs. 1–100 wt%L285, 2–30 wt%Co-L285, 3–60 wt%Co-L285, 4–2 wt%MWCNT/30 wt%Co-L285, 5–5 wt%MWCNT/30 wt%Co-L285, 6–7 wt%MWCNT/30 wt%Co-L285, 7–10 wt%MWCNT/30 wt%Co-L285.

x%MWCNT/30 wt%Co-L285. The MWCNTs exhibit an elongated cylindrical morphology, promoting the formation of a branched conductive network even at low concentrations, whereas the Co particles are nearly spherical, providing additional contact points and polarization centers. In all studied three-phase samples, the Co content is fixed at 30 wt%, while the MWCNT concentration varies from 2 to 10 wt%. For comparison, the graph also includes reference curves for a two-phase composite containing 60 wt%Co, illustrating a limiting case of matrix oversaturation with metallic particles.

In systems without nanotubes (30 wt% and 60 wt%Co), the real part of the permittivity ε' is approximately 4.2 and 4.4, respectively (at 10^6 Hz), and does not show significant frequency dependence. This indicates the dielectric nature of these compositions, where spherical Co particles do not form effective conductive pathways. Increasing the metal content up to 60 wt% does not result in a notable enhancement of ε' but only stabilizes it at a slightly higher level.

As seen in Fig. 4a, the introduction of MWCNTs leads to significant changes in the dielectric behavior. For samples containing 2 wt% and 5 wt%MWCNTs combined with 30 wt% Co, the real permittivity ε' remains close to 9–10, which is only slightly higher than the values observed for the two-component polymer–Co composite. This suggests that, at this concentration, Co does not facilitate efficient electrical bridging between nanotubes and may even hinder the formation of conductive clusters. Only at 7 wt% and especially at 10 wt%MWCNTs does ε' increase sharply – up to ~ 50 and ~ 130 , respectively, at low frequencies. This behavior indicates that the percolation threshold is reached for the nanotube network, which, in the presence of Co, begins to form effective polarization pathways.

Figure 4b presents the frequency dependence of the imaginary part of the dielectric permittivity, ε'' . At a frequency of 1 MHz, ε'' values for the samples without nanotubes are approximately -0.16 for the pure matrix, ~ 0.19 for the composite with 30 wt%Co, and ~ 1.03 for 60 wt%Co. These values indicate purely dielectric behavior, where even a

high content of Co metallic particles does not result in the formation of a conductive network.

The introduction of MWCNTs significantly alters the dielectric response. At 2 wt%MWCNT/30 wt%Co, $\varepsilon' = 0.42$, which remains low; at 5 wt%MWCNTs, it increases to ~ 20.6 , signaling proximity to the percolation threshold. At 7 wt%MWCNTs, ε'' rises sharply to ~ 404 , and at 10 wt%MWCNTs it reaches ~ 1793 . This progression reflects a rapid increase in dielectric losses in the conductive regime following the formation of a nanotube-based network.

Thus, the analysis of the results presented in Fig. 4 confirms that the three-phase system containing cylindrical MWCNTs and spherical Co particles (at a fixed 30 wt%Co) exhibits a clear transition from dielectric to conductive behavior at an MWCNTs concentration of approximately 7 wt%. At 5 wt%, the real permittivity ε' remains at ~ 9 , characteristic of a pre-percolation state. In contrast, at 7 wt%MWCNTs, ε' increases sharply to ~ 60 , and exceeds ~ 150 at 10 wt%, indicating the formation of interconnected conductive pathways.

Compared to the two-phase MWCNTs–L285 composites, the ε' values in the three-phase Co-containing systems are significantly lower in the 2–5 wt%MWCNTs concentration range. This effect is attributed to the fact that spherical Co particles, unlike elongated nanotubes, do not form effective electrical contacts and may partially hinder nanotube-to-nanotube connectivity, thereby reducing interfacial polarization in the early stages of network formation. As a result, a higher MWCNTs concentration is required to achieve the conductive regime in the three-phase system, consistent with observations reported by other researchers in [46].

Figure 5a, b presents the frequency dependences of ε' and ε'' for three-phase composites containing 30 wt%Fe and varying MWCNTs contents (2–10 wt%). For comparison, the dielectric response of the neat L285 matrix and the Fe-filled composite without nanotubes is also shown.

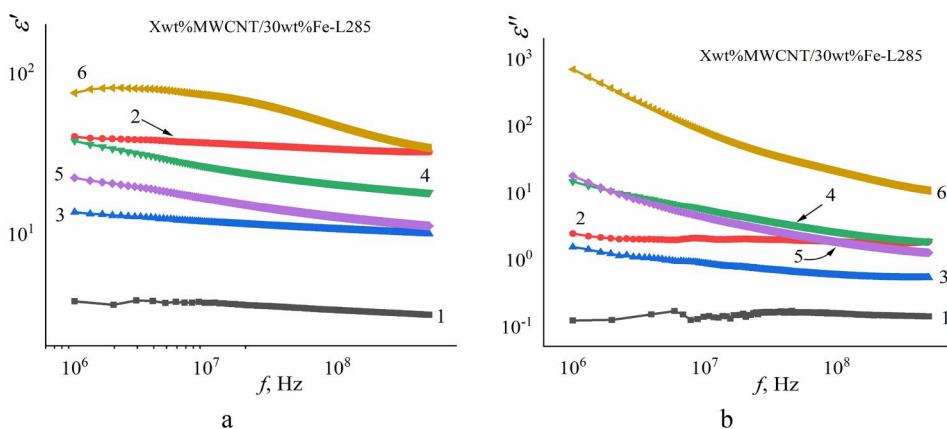


Figure 5. Frequency dependences of the real (ε') (a) and imaginary (ε'') (b) parts of the complex permittivity of MWCNT/30 wt%Fe-L285 epoxy composites with MWCNT contents of 0–10 wt%. 1–100 wt%L285, 2–30 wt%Fe-L285, 3–2 wt%MWCNT/30 wt%Fe-L285, 4–5 wt%MWCNT/30 wt%Fe-L285, 5–7 wt%MWCNT/30 wt%Fe-L285, 6–10 wt%MWCNT/30 wt%Fe-L285.

At a frequency of 1 MHz, the real permittivity ϵ' for the composite containing 30 wt%Fe reaches approximately ~ 40 , indicating a significant contribution of Fe to polarization processes even in the absence of nanotubes. Upon the addition of 2 wt%MWCNTs, ϵ' decreases to ~ 13.6 and ϵ'' to ~ 1.5 , suggesting that, at low concentrations, Fe particles partially isolate the nanotubes and hinder their mutual interaction. At 5 wt%MWCNTs, ϵ' increases to ~ 38 and ϵ'' to ~ 20 , while at 7 wt% both parameters decrease again to ~ 22 and ~ 13 , respectively. The highest values are recorded at 10 wt%MWCNTs: $\epsilon' \approx 75$ and $\epsilon'' \approx 694$, indicating the formation of an efficient conductive network and intensive energy absorption at low frequencies.

Thus, the Fe–MWCNT system exhibits a notable increase in dielectric permittivity and losses only at a nanotube concentration of 10 wt%, whereas at lower concentrations, the micron-sized disc-shaped Fe particles appear to reduce the efficiency of MWCNT interconnection. This effect may be attributed to the geometry of the Fe particles – their limited contact area with the elongated nanotubes and their tendency to partially block percolation pathways.

In comparison with the three-phase Co-based system and the two-phase composite containing 10 wt%MWCNTs, the sample with 30 wt% Co displays the highest values of ϵ' (~ 152) and ϵ'' (~ 1793), whereas the 30 wt%Fe system reaches only $\epsilon' \sim 75$ and $\epsilon'' \sim 694$. These results suggest that Co more effectively activates both conductivity and interfacial polarization. This is explained not only by differences in intrinsic electrical conductivity, but also by particle morphology: spherical Co particles provide more efficient interaction with nanotubes than disc-shaped micron-sized Fe particles.

MWCNTs used in the composites typically possess high aspect ratios, with lengths reaching several micrometers and diameters in the range of ~ 10 – 30 nm, along with an inherent tendency to entangle into flexible thread-like bundles. This morphology is critical for the formation of efficient conductive pathways as the system approaches the effective percolation threshold.

The Fe particles employed in this work exhibit a disc-like shape with lateral dimensions on the order of several micrometers – comparable to the nanotube length but significantly larger than their thickness. This geometry likely prevents the Fe disks from penetrating the MWCNT bundles and may instead locally restrict mobility or separate entangled MWCNT structures, thereby hindering the development of three-dimensional conductive networks.

This behavior explains the observed effect: the addition of Fe particles alone increases ϵ' due to interfacial polarization. However, introducing small amounts of MWCNTs (2–5 wt%) does not lead to any rise in ϵ' or ϵ'' ; the system remains dielectric or even underperforms compared to the Fe-only composite. Only at 10 wt%MWCNTs do the nanotubes likely form interconnected chains that overcome the geometric hindrance imposed by the rigid Fe disks. As a result, the onset of apparent percolation behavior is shifted to higher MWCNT concentrations compared to two-phase systems.

Figure 6a, b illustrates the frequency dependence of the real (ϵ' , a) and imaginary (ϵ'' , b) parts of the complex dielectric permittivity for three-phase composites based on epoxy resin L285 with the addition of 30 wt% Co oxide (Co_3O_4) particles and 2–10 wt%MWCNTs. The data are presented in comparison with the pure matrix (100%L285) and a two-phase system containing 30 wt% Co_3O_4 without MWCNTs.

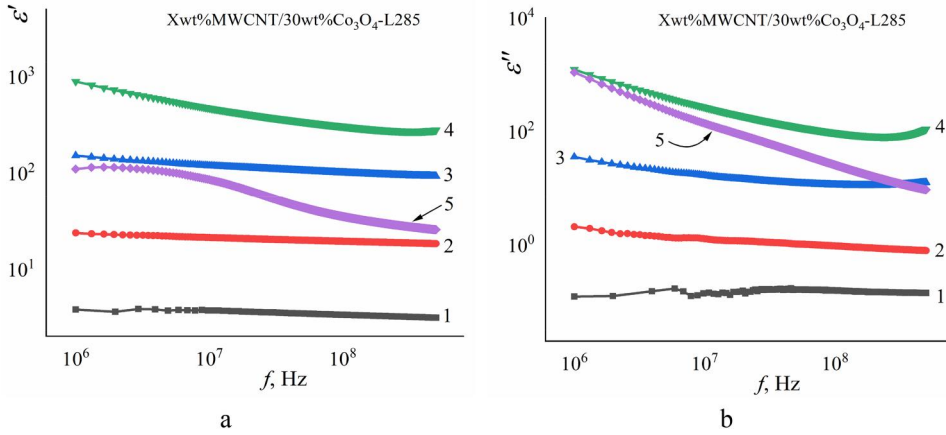


Figure 6. Frequency dependences of the real (ϵ') (a) and imaginary (ϵ'') (b) parts of the complex permittivity of x%MWCNT/30 wt%Co₃O₄-L285 epoxy composites with MWCNT contents of 0–10 wt%. 1–100 wt%L285, 2–30 wt%Co₃O₄-L285, 3–2 wt%MWCNT/30 wt%Co₃O₄-L285, 4–5 wt%MWCNT/30 wt%Co₃O₄-L285, 5–10 wt%MWCNT/30 wt% Co₃O₄-L285.

This system employs nanoscale spherical Co₃O₄ particles (~ 50 nm in size), which are substantially different from the micron-sized fillers used in the previously discussed composites.

As shown in Fig. 6a, the addition of 30 wt%Co₃O₄ increases ϵ' from 3.8 to ~ 24 . With 2 wt%MWCNTs, ϵ' exceeds 100, and at 5 wt% it rises to ~ 900 – a record value among all systems studied. This behavior may be associated with the nanosized Co₃O₄ particles, which may contribute to enhanced polarization processes and the formation of conductive pathways. However, with a further increase in MWCNTs content to 10 wt%, ϵ' decreases to ~ 109 . This reduction may be due to the formation of closed conductive loops that limit polarization charge accumulation.

Figure 6b shows the frequency-dependent behavior of the imaginary part of the dielectric permittivity for the Co₃O₄-based composites. As the MWCNT content increases from 2 wt% to 5 wt%, ϵ'' rises significantly from ~ 35 to ~ 1216 at 1 MHz, indicating strong interfacial polarization and relaxation losses. This behavior suggests that the system is approaching a percolation transition associated with the formation of conductive pathways. A further increase in MWCNT content to 10 wt% results in a decrease of ϵ'' to ~ 1087 , which is likely attributed to reduced polarization efficiency due to the increased conductive contribution and partial field screening by the conductive pathways.

As a result, the Co₃O₄-containing composites approach the percolation transition at lower MWCNT concentrations than the other investigated systems. This enables the achievement of the lowest percolation threshold among all the investigated three-phase systems while maintaining a stable dielectric response over a broad frequency range.

Figure 7a, b presents the concentration-dependent behavior of the real (ϵ' , a) and imaginary (ϵ'' , b) parts of the dielectric permittivity for epoxy composites containing various inorganic fillers (Co, Fe, and Co₃O₄) at a fixed frequency of $f = 10^6$ Hz. In all systems, ϵ' increases with rising MWCNT content. The insets in the graphs show the

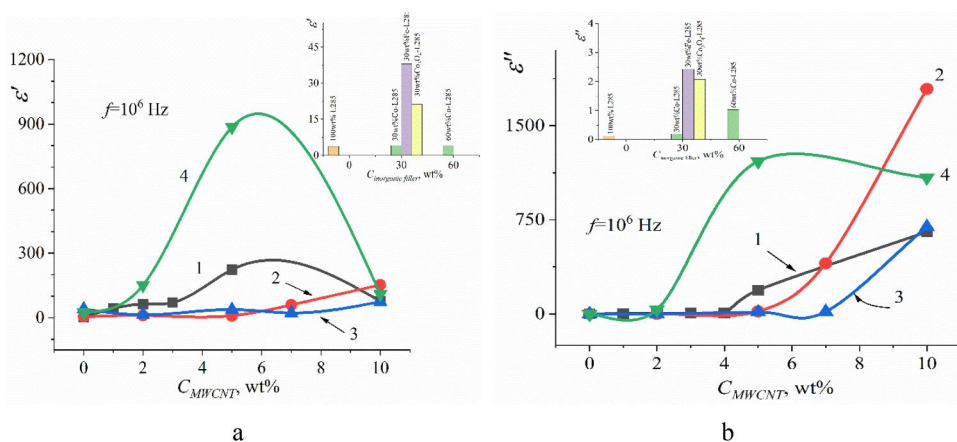


Figure 7. Concentration dependences of the real (ϵ') (a) and imaginary (ϵ'') (b) parts of the complex permittivity of MWCNT/L285 epoxy composites and three-component MWCNT/Co-L285, MWCNT/Fe-L285, and MWCNT/Co₃O₄-L285 systems measured at $f = 10^6$ Hz. The insets show the influence of metallic (Co, Fe) and oxide (Co₃O₄) fillers at different loadings (0, 30, and 60 wt%) on ϵ' and ϵ'' . 1 – Xwt%MWCNT-L285, 2 – Xwt%MWCNT/30 wt%Co-L285, 3 – Xwt%MWCNT/30 wt%Fe-L285, 4 – Xwt%MWCNT/30 wt%Co₃O₄-L285.

dielectric properties of systems containing 30–60 wt% of a single type of inorganic filler without the addition of MWCNTs. These data allow for the evaluation of the intrinsic contribution of Co, Fe, and Co₃O₄ particles to the polarization of the polymer matrix.

Among these, the 30 wt%Fe-L285 system exhibits the highest increase in both ϵ' and ϵ'' in the insets, which can be attributed to the high electrical conductivity of the micron-sized Fe particles. In contrast, Co₃O₄ and Co provide smaller contributions to the permittivity, consistent with their lower electrical conductivity. The lowest values are observed for the neat epoxy matrix L285 ($\epsilon' \approx 4$, $\epsilon'' \approx 0.2$), which serves as the baseline reference system.

In Fig. 7a, which illustrates the concentration dependence of $\epsilon'(C)$, the highest increase in ϵ' is observed for the three-phase MWCNT/30 wt%Co₃O₄-L285 system. In this composite, at 5–7 wt%MWCNTs, ϵ' exceeds 900 – significantly higher than in any other system studied. This indicates highly efficient electric energy storage due to active charge displacement and orientation at the phase boundaries. Such behavior is attributed to the nanoscale size (~ 50 nm) and spherical morphology of the Co₃O₄ particles, which are uniformly distributed within the polymer matrix and due to their nanosized dimensions and more homogeneous distribution within the polymer matrix. The significantly higher ϵ' and ϵ'' values of the MWCNT/Co₃O₄ composites compared to the other investigated MWCNT/magnetic filler systems are attributed to enhanced interfacial polarization arising from the large number of interfacial boundaries formed by nano-sized Co₃O₄ particles and MWCNTs with high specific surface area S_m (see Table 1).

For comparison, in the two-phase MWCNT-L285 composite, the maximum value of ϵ' reaches approximately 280 at 7 wt%MWCNTs. This increase is primarily attributed to the formation of conductive pathways by the nanotubes themselves, a typical feature of the percolation effect. Since this system lacks a second functional filler, its structure

develops spontaneously, without morphological influence from additional particles. Consequently, the dielectric properties are limited by the intrinsic ability of the nanotubes to form a percolated conductive network.

In contrast, the Co- and Fe-containing composites exhibit lower ϵ' values (~ 100 and ~ 60 , respectively, at 10 wt%MWCNTs), which is attributed to the larger particle size and less favorable morphology of the micron-sized fillers. Fig. 7b, which illustrates the concentration dependence of the imaginary part of the dielectric permittivity ϵ'' , also reveals a characteristic increase in dielectric losses with rising MWCNT content across all systems. The highest ϵ'' values are observed in the three-phase composite with 30 wt%Co, where ϵ'' exceeds 1700 at 10 wt%MWCNTs, indicating the dominance of conductive losses in the over-percolated regime. In contrast, the Co_3O_4 -based system shows a much lower ϵ'' value of approximately 800 at the same MWCNT concentration. This suggests that Co_3O_4 contributes to lower dielectric losses while maintaining high ϵ' values, likely due to the nanosized character and lower intrinsic conductivity of the oxide particles. Compared to the Fe-containing system and the two-phase composite, the Co_3O_4 -containing composites exhibit higher ϵ' values while maintaining comparatively lower dielectric losses.

A comparison of all investigated systems indicates that the incorporation of nanosized Co_3O_4 particles leads to the highest dielectric response and the earliest onset of conductive network formation among the studied composites. This is evidenced by the steep increase in permittivity observed at just 3–5 wt%MWCNTs. In contrast, spherical Co particles ($\sim 1.5\ \mu\text{m}$) and disc-shaped Fe particles ($\sim 3\text{--}7\ \mu\text{m}$), due to their larger size, are less favorable for conductive pathway formation, thereby increasing the percolation threshold. As a result, in Co- and Fe-containing composites, electrical conductivity emerges at higher MWCNT concentrations and is accompanied by greater dielectric losses (ϵ'').

Thus, the results in Fig. 7a, b confirm the crucial role of inorganic filler morphology in shaping the dielectric properties of three-phase composites. Specifically, the nanoscale Co_3O_4 particles are associated with enhanced polarization capability and promote earlier conductive network formation, resulting in the highest dielectric response among the investigated composites [41,44].

The high ϵ' and ϵ'' values observed for composites with high MWCNT concentrations are associated with enhanced interfacial polarization processes and increased conductive contributions in systems approaching the percolation transition. Similar dielectric behavior, including strong permittivity enhancement and increased dielectric losses near the percolation transition, has previously been reported for epoxy-based nanocarbon composites in the radio-frequency range [47,48], as well as for CNT/epoxy metacomposites exhibiting conductive network formation and large dielectric responses [49].

3.2. Frequency dependence of AC electrical conductivity in epoxy composites with nanocarbon and inorganic fillers

The frequency dependence of AC electrical conductivity was analyzed for x%MWCNT–L285 and x%MWCNT/30 wt%filler–L285 composites, where x varied from 2 to 10 wt%. The analysis of the frequency dependence of AC electrical conductivity, $\sigma_{AC}(f)$, for all

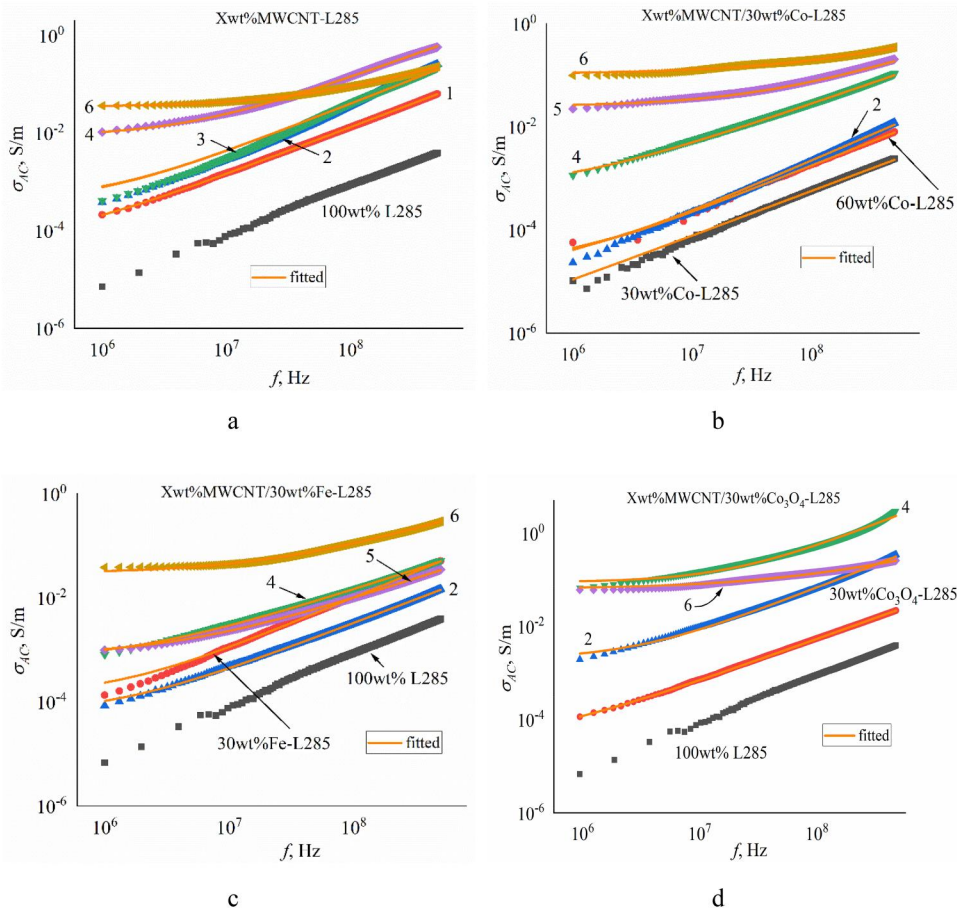


Figure 8. Frequency dependences of the AC electrical conductivity (σ_{AC}) of (a) MWCNT/L285 epoxy composites and three-component (b) MWCNT/30 wt%Co-L285, (c) MWCNT/30 wt%Fe-L285, and (d) MWCNT/30 wt%Co₃O₄-L285 systems at different MWCNT contents. Solid lines represent fits to the power-law model, Equation (4). 1 – $X_{MWCNT} = 1\text{wt}\%$, 2 – $X_{MWCNT} = 2\text{wt}\%$, 3 – $X_{MWCNT} = 3\text{wt}\%$, 4 – $X_{MWCNT} = 5\text{wt}\%$, 5 – $X_{MWCNT} = 7\text{wt}\%$, 6 – $X_{MWCNT} = 10\text{wt}\%$.

investigated epoxy composites (Fig. 8), together with the fitting parameters summarized in Table 2, provides a consistent understanding of the charge transport mechanisms across systems with different phase compositions.

Figure 8a–d presents the frequency dependences of the AC electrical conductivity, $\sigma_{AC}(f)$, for (a) MWCNT/L285 and three-component (b) MWCNT/30 wt%Co-L285, (c) MWCNT/30 wt%Fe-L285, and (d) MWCNT/30 wt%Co₃O₄-L285, measured at various MWCNT loadings in the frequency range of 10^6 – $5 \cdot 10^8$ Hz. The AC electrical conductivity behavior in disordered polymer composites containing conductive fillers is a universal feature of such systems and reflects changes in charge transport mechanisms with increasing frequency and the degree of structural connectivity within the conductive network. Within the framework of dielectric spectroscopy, the AC conductivity was determined from the imaginary part of the complex dielectric permittivity using the relation [50,51]:

Table 2. Parameters A , u , and relaxation time τ_c , and critical frequency f_c for nanocarbon-inorganic filler-epoxy composites

System	σ_{DC} , S/m	A	u	f_c , Hz	τ_c , s
Inorganic filler L285					
30%Co-L285	$5.3 \cdot 10^{-7}$	$6.5 \cdot 10^{-11}$	0.9	$3.3 \cdot 10^4$	$4.9 \cdot 10^{-6}$
60%Co-L285	$1.6 \cdot 10^{-5}$	$7.0 \cdot 10^{-11}$	0.9	$6.3 \cdot 10^5$	$2.5 \cdot 10^{-7}$
30%Fe-L285	$1.4 \cdot 10^{-4}$	$1.0 \cdot 10^{-10}$	1.0	$1.3 \cdot 10^6$	$1.2 \cdot 10^{-7}$
30%Co ₃ O ₄ -L285	$3.5 \cdot 10^{-5}$	$3.3 \cdot 10^{-10}$	0.9	$3.9 \cdot 10^5$	$4.1 \cdot 10^{-7}$
Nanocarbon-L285					
C_{MWCNTS} , wt%	σ_{DC} , S/m	A	u	f_c , Hz	τ_c , s
1%	$5.6 \cdot 10^{-5}$	$2.4 \cdot 10^{-10}$	1.0	$3.5 \cdot 10^5$	$4.5 \cdot 10^{-7}$
2%	$3.8 \cdot 10^{-4}$	$4.0 \cdot 10^{-10}$	1.0	$9.6 \cdot 10^5$	$1.7 \cdot 10^{-7}$
3%	$4.1 \cdot 10^{-5}$	$4.0 \cdot 10^{-10}$	1.0	$1.0 \cdot 10^6$	$1.6 \cdot 10^{-7}$
5%	$8.6 \cdot 10^{-3}$	$5.9 \cdot 10^{-9}$	0.9	$5.0 \cdot 10^6$	$3.2 \cdot 10^{-8}$
10%	$3.4 \cdot 10^{-2}$	$4.0 \cdot 10^{-8}$	0.8	$5.2 \cdot 10^7$	$3.1 \cdot 10^{-9}$
Nanocarbon/inorganic filler-L285					
MWCNT/30 wt%Co-L285					
2%	$2.3 \cdot 10^{-5}$	$2.2 \cdot 10^{-11}$	1.0	$1.1 \cdot 10^6$	$1.5 \cdot 10^{-7}$
5%	$6.2 \cdot 10^{-4}$	$1.1 \cdot 10^{-8}$	0.8	$8.3 \cdot 10^5$	$1.9 \cdot 10^{-7}$
7%	$2.5 \cdot 10^{-2}$	$7.3 \cdot 10^{-8}$	0.7	$3.4 \cdot 10^7$	$4.7 \cdot 10^{-9}$
10%	$1.1 \cdot 10^{-1}$	$1.4 \cdot 10^{-6}$	0.6	$1.3 \cdot 10^8$	$1.2 \cdot 10^{-9}$
MWCNT/30 wt%Fe-L285					
2%	$5.8 \cdot 10^{-5}$	$1.4 \cdot 10^{-10}$	0.9	$1.3 \cdot 10^6$	$1.3 \cdot 10^{-7}$
5%	$6.1 \cdot 10^{-4}$	$7.9 \cdot 10^{-9}$	0.8	$1.8 \cdot 10^6$	$8.9 \cdot 10^{-8}$
7%	$7.9 \cdot 10^{-4}$	$4.8 \cdot 10^{-9}$	0.8	$4.3 \cdot 10^6$	$3.7 \cdot 10^{-8}$
10%	$3.0 \cdot 10^{-2}$	$1.3 \cdot 10^{-7}$	0.7	$2.5 \cdot 10^7$	$6.4 \cdot 10^{-9}$
MWCNT/30 wt%Co ₃ O ₄ -L285					
2%	$2.0 \cdot 10^{-3}$	$6.3 \cdot 10^{-10}$	1.0	$3.2 \cdot 10^6$	$5.0 \cdot 10^{-8}$
5%	$9.0 \cdot 10^{-2}$	$4.5 \cdot 10^{-9}$	1.0	$2.0 \cdot 10^7$	$8.0 \cdot 10^{-9}$
10%	$6.4 \cdot 10^{-2}$	$1.2 \cdot 10^{-6}$	0.6	$7.7 \cdot 10^7$	$2.1 \cdot 10^{-9}$

$$\sigma_{AC}(f) = 2\pi f \varepsilon_0 \varepsilon''(f) \quad (3)$$

For all studied systems, σ_{AC} exhibits a characteristic two-region behavior: a low-frequency plateau with weak frequency dependence, followed by a power-law increase above a critical frequency f_c . With increasing MWCNT content, the overall conductivity increases and the low-frequency plateau becomes more pronounced, reflecting the progressive formation of interconnected conductive pathways.

The experimental $\sigma_{AC}(f)$ dependences were fitted using the universal power-law model [52]:

$$\sigma_{AC}(f) = \sigma_{DC} + A f^u \quad (4)$$

where σ_{DC} is the DC conductivity, A is a pre-exponential factor related to the density of conductive pathways, and u is the frequency exponent associated with the charge transport mechanism. The calculated curves obtained using Jonscher's power law demonstrate satisfactory agreement with the experimental frequency dependences of AC conductivity for the investigated composites. The exponent u varies from values close to $u \approx 1$, typically associated with hopping-related conduction behavior, to sublinear values in the range $u \approx 0.6$ – 0.8 . Such a decrease in u may indicate an increasing contribution of delocalized transport along interconnected conductive pathways and is a typical signature of percolative systems [52,53]. This relationship, commonly referred to as

Jonscher's universal power law or the universal dielectric response, arises from the statistical distribution of energy barriers and hopping distances in disordered media [52,54–56]. The solid lines in Fig. 8 represent the fitting curves obtained using Equation (4), showing good agreement with the experimental data across the entire frequency range.

At low frequencies, the dominant contribution arises from σ_{DC} , which corresponds to charge transport along extended conductive paths or a percolation network. As the frequency increases, the contribution from the frequency-dependent term Af^u becomes more significant, reflecting localized transport processes – such as hopping or tunneling of carriers between neighboring conductive domains. The critical frequency f_c , marking the transition between these two regimes, is associated with a characteristic relaxation time.

$$\tau_c = \frac{1}{2\pi f_c} \quad (5)$$

The onset of the power-law region defines the critical frequency f_c , which corresponds to the condition where the dispersive term becomes comparable to the DC component. The frequency dependence of σ_{AC} can also be described by the symmetric hopping model (also known as the random free-energy barrier model) proposed by Dyre, expressed as:

$$\sigma_{AC}(\omega) = \sigma_{DC} \left[\frac{i\omega\tau_c}{\ln(1 + i\omega\tau_c)} \right] \quad (6)$$

where $\omega = 2\pi f$. This model directly links the critical frequency to the characteristic time of hopping or quasi-dispersive motion of charge carriers and has been successfully applied to the analysis of AC conductivity in composites, amorphous semiconductors, and ionic conductors [52,54]. The solid lines in Fig. 8 represent the fitting curves obtained using Equations (4) and (6), demonstrating good agreement with the experimental data over the entire frequency range.

As follows from Fig. 8 and Table 2, increasing the MWCNT content leads to higher values of σ_{DC} , a decrease in the relaxation time τ_c , and a shift of the critical frequency f_c toward higher values. The pristine epoxy polymer L285 is characterized by very low DC conductivity ($\sigma_{DC} \sim 10^{-12}$ S/m), indicate that charge transport is dominated by localized processes associated with isolated polarization mechanisms.

The introduction of inorganic fillers (Co, Fe, and Co_3O_4) results in a moderate increase in σ_{DC} and a decrease in τ_c , suggesting the formation of additional charge transport pathways associated with interfacial polarization and partial involvement of metallic or oxide particles in conductivity. At the same time, the power-law exponent u remains in the range of 0.9–1.0, consistent with a hopping conduction mechanism in disordered media.

For the MWCNT/L285 system, increasing the nanotube concentration results in a sharp rise in σ_{DC} by several orders of magnitude and a systematic decrease in the relaxation time τ_c . Simultaneously, the critical frequency f_c shifts toward higher values, indicating an expansion of the low-frequency $\sigma_{AC}(f)$ plateau. This behavior is consistent with a percolation-like transition: as the MWCNT content increases, interconnected conductive pathways are likely formed, contributing to a more delocalized charge

transport mechanism. The decrease in the exponent u from ~ 1.0 to ~ 0.8 at high MWCNT contents suggests a gradual transition from a purely hopping mechanism to a mixed regime where the contribution of diffusion-assisted transport becomes more significant. Diffusion-assisted charge transport may emerge after the formation of interconnected conductive pathways, where carriers propagate along interconnected conductive pathways with a reduced contribution from thermally activated hopping between localized states. In this regime, the conductivity acquires a more delocalized nature, which is manifested by the reduction in the exponent u and the shift of the critical frequency f_c toward the high-frequency region [52,53,55].

Three-component epoxy composites containing MWCNTs and inorganic fillers (Co, Fe, or Co_3O_4) exhibit markedly different electrical behaviors, governed by the morphology, particle size, and spatial distribution of the inorganic phase, as well as its interaction with the MWCNT conductive network. For MWCNT/30 wt%Co-L285 composites, increasing the MWCNT content leads to a pronounced rise in DC electrical conductivity – from $\sigma_{DC} = 2.3 \cdot 10^{-5}$ S/m at 2 wt%MWCNT to $\sigma_{DC} = 1.1 \cdot 10^{-1}$ S/m at 10 wt%MWCNT. This increase is accompanied by a gradual reduction in the frequency exponent u from 1.0 to 0.6 and a decrease in the relaxation time τ_c to $1.2 \cdot 10^{-9}$ s, suggesting a transition from localized hopping conduction toward diffusion-assisted transport behavior within a well-developed conductive network. The high critical frequencies ($f_c \sim 10^8$ Hz) suggest a stable conductive regime over a broad frequency range.

In MWCNT/30 wt%Fe-L285 composites, the formation of an efficient conductive network is less pronounced. At 10 wt%MWCNT, the DC conductivity reaches $\sigma_{DC} = 3.0 \cdot 10^{-2}$ S/m, which is significantly lower than in the Co-containing system. The exponent u remains in the range of 0.7–0.9, while the relaxation times τ_c are relatively longer (10^{-8} – 10^{-9} s), indicating a predominant contribution from hopping conduction. This behavior is attributed to the micron-sized, disk-like Fe particles, which are less effective in promoting continuous CNT-CNT contacts and hinder the formation of percolated conductive pathways.

The most pronounced enhancement in electrical conductivity is observed for MWCNT/30 wt% Co_3O_4 -L285 composites. Already at 2 wt%MWCNT, the DC conductivity reaches $\sigma_{DC} = 2.0 \cdot 10^{-3}$ S/m, exceeding that of Co- and Fe-based systems by 1–2 orders of magnitude at comparable nanotube concentrations. At 5 wt%MWCNT, σ_{DC} increases to $9.0 \cdot 10^{-2}$ S/m, while the frequency exponent u remains close to unity, indicating the early-stage formation of an efficient conductive network. At 10 wt%MWCNT, a reduction of u to 0.6, along with minimal relaxation times ($\tau_c \sim 2 \cdot 10^{-9}$ s) suggests an increasing contribution of diffusion-assisted charge transport. This behavior may be associated with the nanosized Co_3O_4 particles, which can contribute to more effective formation of conductive pathways and enhanced interparticle connectivity within the composite system.

The observed trends are consistent with the power-law description of AC conductivity (Equation (4)) and the relaxation -based transport model (Equation (6)). The fitted parameters A , u , τ_c , and f_c , summarized in Table 1, quantitatively support these trends and indicate that charge transport in the studied composites evolves from hopping-dominated conduction at low filler contents ($u \approx 1.0$) toward a percolation-assisted,

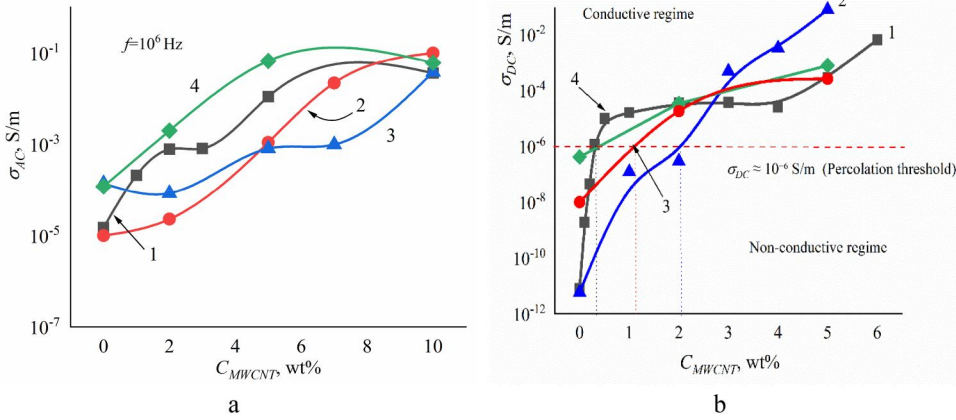


Figure 9. Dependence of electrical conductivity on MWCNT content in epoxy-based composites: (a) AC conductivity σ_{AC} and (b) DC conductivity σ_{DC} , illustrating the percolation behavior in MWCNT/L285 composites with different inorganic fillers. 1 – Xwt%MWCNT-L285, 2 – Xwt%MWCNT/30 wt%Co-L285, 3 – Xwt%MWCNT/30 wt%Fe-L285, 4 – Xwt%MWCNT/30 wt%Co₃O₄-L285.

diffusion-controlled regime at higher loadings, where the exponent decreases to $u \approx 0.6$ – 0.7 .

Figure 9a, b presents the dependencies of AC conductivity and DC conductivity on the concentration of MWCNT C_{MWCNTs} for epoxy composites based on L285 containing different inorganic fillers (Co, Fe, and Co₃O₄). As shown in Fig. 9a, σ_{AC} increases with increasing MWCNT content, reflecting the progressive formation of a conductive percolation network. This behavior is typical of polymer nanocomposites with carbon-based fillers and aligns with the classical framework of percolation theory [57]. The highest values of σ_{AC} are observed for the MWCNT/30 wt%Co₃O₄-L285 composites, suggesting the most efficient formation of conductive pathways in the presence of Co oxide nanoparticles.

The DC conductivity trends (Fig. 9b) clearly demonstrate the percolative nature of charge transport. All systems exhibit a transition from an insulating to a conductive regime upon reaching a threshold conductivity of approximately $\sigma_{DC} \approx 10^{-6}$ S/m, which is commonly considered the criterion for the formation of a continuous conductive network in polymer composites [58,59]. The percolation behavior of composite systems containing hybrid MWCNT/magnetic fillers has been discussed in greater detail in our previous studies [38,60,61]. For the MWCNT/30 wt%Co₃O₄-L285 systems, the percolation threshold is achieved at lower MWCNT concentrations than in the Co- and Fe-based composites as well as in the two-component MWCNT-L285 system. This behavior suggests that nanosized Co₃O₄ particles facilitate earlier conductive network formation within the composite.

The combined analysis of AC and DC data shows that the increase in σ_{AC} at a fixed frequency correlates well with the behavior of σ_{DC} , suggesting the dominant contribution of electronic transport mechanisms associated with the formation of branched conductive pathways. In the concentration range above the percolation threshold, the contribution of diffusive electron transport increases, while the role of hopping conductivity gradually diminishes, which is characteristic of densely interconnected conductive networks in polymer nanocomposites [62].

Therefore, the analysis of the dependencies shown in Fig. 9 demonstrates that the presence of Co_3O_4 in three-component epoxy composites significantly influences the formation of the conductive network. In particular, the addition of Co_3O_4 promotes a reduction in the effective percolation threshold and results in higher AC and DC conductivity values compared to two-component systems and composites with metallic fillers.

3.3. Frequency dependence of the real (μ') and imaginary (μ'') parts of the complex magnetic permeability in MWCNT/epoxy composites containing Co, Fe, and Co_3O_4 fillers (1–500 MHz)

Figure 10a–d presents the frequency dependences of the real μ' and imaginary μ'' parts of the complex magnetic permeability for the two-component MWCNT/L285 composite and the three-component composites MWCNT/Co-L285, MWCNT/Fe-L285, and

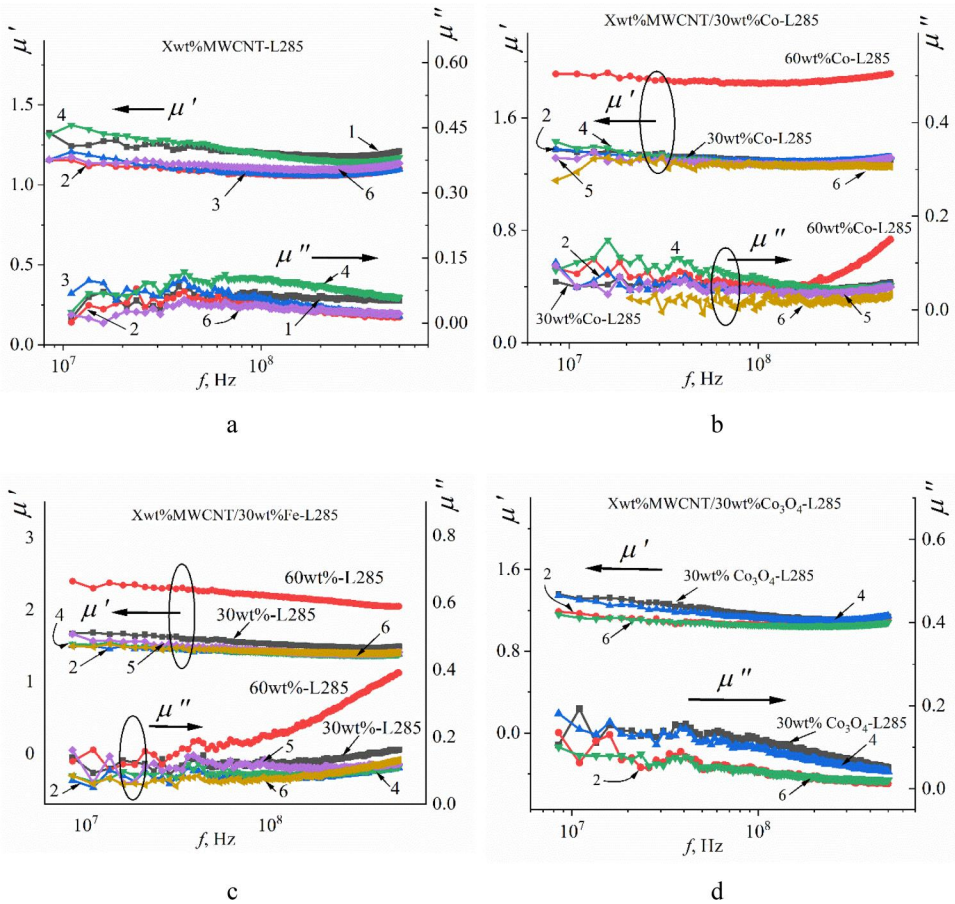


Figure 10. Frequency dependences of the real (μ') and imaginary (μ'') parts of the complex magnetic permeability of (a) MWCNT/L285, (b) MWCNT/Co-L285, (c) MWCNT/Fe-L285, and (d) MWCNT/ Co_3O_4 -L285 epoxy composites at different filler loadings. 1 – $X_{\text{MWCNT}} = 0\text{wt}\%$, 2 – $X_{\text{MWCNT}} = 1\text{wt}\%$, 3 – $X_{\text{MWCNT}} = 2\text{wt}\%$, 4 – $X_{\text{MWCNT}} = 3\text{wt}\%$, 5 – $X_{\text{MWCNT}} = 5\text{wt}\%$, 6 – $X_{\text{MWCNT}} = 7\text{wt}\%$, 7 – $X_{\text{MWCNT}} = 10\text{wt}\%$.

MWCNT/Co₃O₄-L285 in the frequency range of approximately $\sim 10^7$ – $5 \cdot 10^8$ Hz at various filler concentrations.

For all studied systems, the real part of the magnetic permeability (μ') exhibits a weakly decreasing frequency dispersion, which is typical for composites containing magnetic fillers in the high-frequency range. In the two-component MWCNT/L285 composite (Fig. 10a), the values of μ' lie in the range of approximately 1.1–1.3. Three and decrease only slightly with increasing frequency. The slight enhancement of μ' above unity observed for MWCNT/L285 composites is not related to the presence of an intrinsic magnetic phase, as carbon nanotubes are non-magnetic materials. In the literature, the magnetic permeability of MWCNTs/polymer composites is typically assumed to be unity, and deviations from this value are attributed to electrodynamic effects associated with the formation of conductive networks and the induction of eddy currents in the radiofrequency range [63–65]. Specifically, near the percolation threshold, the increase in σ_{AC} and ε' promotes the emergence of inductive contributions to the electromagnetic response, which may manifest as an effective increase in μ' without involving intrinsic magnetic mechanisms.

For MWCNT/Co-L285 and MWCNT/Fe-L285 composites (Fig. 10b, c), the real part of the magnetic permeability (μ') is significantly higher, reaching values of ~ 1.6 – 2.4 , especially at elevated contents of metallic filler (30–60 wt%). This behavior is attributed to the ferromagnetic nature of Co and Fe and is associated with the contribution of domain and bulk magnetic polarization, which is characteristic of metal-polymer composites containing ferromagnetic fillers. Similar mechanisms of magnetic response formation, governed by the domain structure and magnetic polarization of Fe, Co, and Ni particles embedded in polymer matrices, have been discussed in detail in [66]. The weak decrease in μ' with increasing frequency may be related to the limited mobility of domain walls and the gradual transition to a magnetic relaxation regime, which is typical for ferromagnetic composite systems.

In MWCNT/Co₃O₄-L285 systems (Fig. 10d), the real part of the magnetic permeability remains moderate (~ 1.0 – 1.3) and more stable across the frequency range. This behavior is consistent with the antiferromagnetic/paramagnetic nature of Co₃O₄, as well as the nanoscale dimensions of the particles, which promote surface spin polarization without the development of classical domain structures.

The imaginary component (μ'') reflects magnetic losses associated with relaxation and resonance processes. For MWCNT/L285, μ'' remains low (≤ 0.2) and only weakly frequency-dependent, indicating negligible magnetic losses. Composites containing Co and Fe exhibit significantly higher μ'' , particularly in samples with 60 wt% metal. The increase in μ'' at high frequencies is attributed to intrinsic ferromagnetic resonance and eddy current effects within the metallic particles, which become more pronounced as the composite's electrical conductivity increases. In contrast, μ'' in MWCNT/Co₃O₄-L285 remains low or even decreases with increasing frequency, indicating suppression of eddy currents due to the oxide nature of Co₃O₄ and improved nanoparticle dispersion within the polymer matrix. This behavior is favorable for minimizing magnetic losses in the high-frequency range.

The obtained magnetic properties are in good agreement with the previously discussed frequency- and concentration-dependent trends in electrical conductivity and

dielectric permittivity. In particular, composites containing Co_3O_4 combine enhanced electrical conductivity (due to a reduced percolation threshold of the CNT network) with relatively low magnetic losses; meanwhile, systems with Fe and Co exhibit a stronger magnetic response but also higher losses, attributed to the ferromagnetic nature of the metals and increased eddy current effects.

Thus, the morphology and magnetic nature of the inorganic filler play a key role in shaping the overall electromagnetic response of the three-phase epoxy composites.

4. Conclusion

1. This study investigates the frequency-dependent complex dielectric permittivity (ϵ' and ϵ'') of epoxy composites filled with multiwalled carbon nanotubes and inorganic particles of different morphologies: spherical Co_3O_4 nanoparticles (Co_3O_4 , ~ 50 nm), spherical micron-sized Co particles (Co, < 1.5 μm), and disk-shaped Fe particles (Fe, ~ 3 – 7 μm). It was found that the morphology of the secondary filler critically influences the formation of both the real and imaginary components of the dielectric response. The MWCNT/ Co_3O_4 -filled system exhibited the highest ϵ' values (up to 900) and ϵ'' values (up to 1200) at 5–7 wt%MWCNTs, indicating enhanced interfacial polarization associated with the formation of a large number of interfacial boundaries (due to the use of nanosized Co_3O_4 particles and MWCNTs) and an increase in dielectric losses in the radio-frequency range. In contrast, the use of micron-sized Co and Fe filler particles in combination with MWCNTs results in the formation of a smaller number of interfacial boundaries and, consequently, a lower level of interfacial polarization (ϵ') and dielectric losses (ϵ'') in the pre-percolation region.
2. The electrical behavior of the composites was further evaluated through the frequency-dependent AC conductivity. The results revealed a gradual transition from hopping-dominated charge transport at low MWCNT concentrations to a diffusion-assisted regime at higher loadings, as evidenced by increasing conductivity and decreasing relaxation times. The best electrical performance was observed for the Co_3O_4 -based composites, which exhibited the highest DC conductivity and the shortest relaxation times. This behavior may be associated with a more efficient development of the conductive network due to the large number of potential contacts formed between nanosized Co_3O_4 particles as well as between Co_3O_4 particles and MWCNTs. In contrast, systems containing micron-sized Co and Fe particles demonstrated lower conductivity and predominantly hopping-type transport, due to less favorable conditions for conductive pathway formation.
3. The introduction of ferromagnetic fillers such as Co and Fe was found to increase the real part of magnetic permeability (μ' up to 2.4) and lead to significant magnetic losses (μ''), particularly at high filler loadings (60wt%). In contrast, nanoscale Co_3O_4 particles, owing to their paramagnetic nature and nanosized dimensions of the particles, ensured stable μ' values (~ 1.1 – 1.3) and minimal magnetic losses, which is favorable for applications in the high-frequency range.

4. It was found that the structural and morphological characteristics of the inorganic filler – particularly particle size, shape, and aggregation tendency – play a critical role in determining the dielectric, conductive, and magnetic properties of the three-phase composites. The most pronounced electromagnetic response was observed in systems incorporating nanoscale cobalt oxide (Co_3O_4), highlighting their strong potential for the development of next-generation multifunctional materials for high-frequency electronics, electromagnetic absorbers, and sensor applications.

Author contributions

CRedit: **Yulia Perets**: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing; **Lyudmila Vovchenko**: Conceptualization, Formal analysis, Methodology, Project administration, Resources, Validation, Writing – review & editing; **Oleg Olikh**: Conceptualization, Formal analysis, Project administration, Resources, Writing – review & editing; **Dmytro Zaiats**: Data curation, Formal analysis, Investigation, Writing – original draft; **Olena Yakovenko**: Data curation, Methodology, Resources, Writing – review & editing; **Lyudmila Matzui**: Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – review & editing; **Iryna Ovsiienko**: Formal analysis, Validation, Visualization, Writing – review & editing; **Volodymyr Matsuy**: Data curation, Methodology, Resources, Writing – review & editing.

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